

matures, expectations of a new generation of employees from their workplaces have also increased. We see employers adopting new technologies that allow them to harness the best from their teams and, in turn, boost their engagement and productivity. These developments have led to the emergence of smart and connected offices, where the IoT drives new ways to collaborate, innovate and socialise,” highlights Joshi.

Elements of a smart workplace

According to Mathews, “There are two key elements—direct employee interface and backend interface—for employee-facing technologies that can provide for flexibility and reduce the time engaged in resolving non-core activities (book a space when you want with room booking tech, locate teams easily for an effective collaboration, tech-enabled vertical transportation, parking, cafeteria, etc).

“For backend interface, tech-enabled physical infrastructure around lighting, HVACs, AVs, rest room service alerts and more help resolve issues proactively, and create business intelligence for predictive analysis to reduce reactive support and accurate business planning.”

Lighting. It is a crucial element of the smart workplace. It largely impacts energy consumption and the Circadian rhythm of employees. Joshi, while explaining the impact of connected lighting, says, “Connected lighting can offer various insights about a building by integrating seamlessly with the IT systems in the building, demonstrating how dynamic office spaces can become smarter and more efficient than ever before.

“Each luminaire in the system sends and receives data, serving as a pathway to deliver value to employees, chief information officers and facilities managers. Managers can track occupancy patterns, changes in temperature, light levels and more, while employees can personalise the lighting around their desks. The system allows maximum visibility and better control. It allows offices to reduce energy consumption. While typically, lighting is responsible for 40 per cent of a building’s electricity use,

IoT-enabled lighting can dramatically reduce this energy consumption.”

Over the past few years there has been an explosion of new mobile devices, primarily smartphones and tablets, into the market. As these devices have become a part of everyday life, expectations of being always connected have extended to workplace technology, too.

Satya Raj, principal, Raj Consultants, shares, “In an artificial intelligence (AI)-enhanced workspace, such as an office or conference room, the environment adapts to your needs and preferences, automatically adjusting the lighting, window tint, seat and desk height, and temperature in anticipation of your arrival and time of day.

“Using information such as a scheduled meeting in your calendar, sound of your voice, a gesture or detection of an implanted RIF chip, the entire room automatically changes to suit your personal likes and learns when new adjustments are needed. The sensors, fitted under desktops and tables, help understand space usage and improve workplace utilisation.”

The rise of both information and the Internet has allowed companies to streamline, automate and outsource much of their structure and process-oriented work.

Cloud. Predicting a major role for the cloud has been an evergreen prediction for the past few years, because it is simple reality. Cloud computing and cloud SaaS have continued to grow and permeate our technological DNA in and out of office.

Virtual assistants. AI is becoming ever-present in our working life, even if it is often happening in the background. AI offers huge opportunities for optimisation and decision making. Simple automated tasks that are used to power or augment human activities are becoming *de rigueur*.

The era of video. Better platforms, workplace technology refreshes and improved connections allow us to collaborate and conference over video from anywhere, breaking down the barriers to global productivity. Workplace training has got a significant facelift through interactive video content that can be viewed at any time and any number of times with a little

more than a connection and a smart device.

Future workplace. Visions of a futuristic workplace are becoming a reality with advances in AI, sophisticated displays and app technology. A smart workplace can leverage increased capabilities in data recording to assign things like parking spaces and desk real estate.

Signage and lighting can direct employees to where they need to go, show meetings and calendar appointments, and seamlessly integrate with behind-the-scenes technology to provide a smoother office experience.

The workplace will go through a dramatic transformation in the next few years, driven by changes in employee demographics, new devices and the changing nature of work. Convergence of these trends creates a unique opportunity for companies.

The time is now for CRE, HR and IT leaders to come together to create the next-generation workplace. To accomplish this, these leaders will need to effectively marry workspace design, technology and workplace policy, creating a workplace that is flexible, adaptable and engaging.

This journey will not be easy. There are plenty of organisational, technical and cultural hurdles that can derail an effort of this size and magnitude. But, those who are successful will propel their companies into a new era of efficiency, innovation and profitability.

Challenges

There are various limiting factors to the decision to transition into a smart workplace. Doing it partly for some necessary infrastructural areas or transforming the entire workplace is one such consideration. Relying on a single consult or reaching out to multiple domain specialist consultants is another.

Vinod Mathews emphasises on a consulting domain expert for building a smart workplace in India. He says, “Going to a single vendor for a complete solution is beneficial when it is a greenfield project, where one can work from the ground up. However, in India, we are yet to see players who have this kind of scale. As such,